Training Leaves Traces: Model Lineage Verification for Open-Weight Language Models

Anonymous ACL submission

Abstract

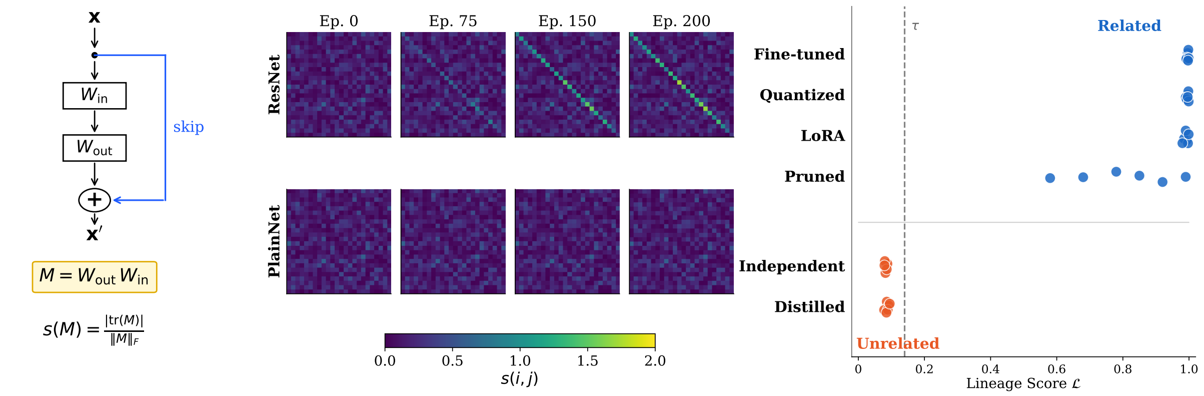
Open-weight models are frequently reused through fine-tuning, quantization, pruning, and merging, but their ancestry is rarely documented in a reliable way. Hashes stop working once weights change, metadata may be incomplete or misleading, and watermarks help only when they are inserted before release. We therefore ask a retrospective question: **given two checkpoints with compatible residual architectures, can we determine whether they share weight ancestry?**

Trained residual blocks exhibit a measurable structure: when their input and output projections are paired correctly, the resulting products show strong normalized trace concentration. This pattern is absent at initialization and does not emerge in matched networks without skip connections. We use this structure to recover projection correspondences, then remove the generic identity-aligned component and compare the remaining centered residual signatures to obtain a model-level lineage score. Across 9 model families spanning language, vision, and speech - including GPT-2, LLaMA-2, ViT, Whisper, and ResNets - the method recovers the correct projection pairings with 80–100% accuracy. We evaluate checkpoint-level lineage verification on controlled residual MLP, ResNet, and GPT-2-style language model benchmarks, as well as public LLaMA-2 checkpoints. Checkpoints produced through fine-tuning, pruning, quantization, and LoRA merging remain close to their ancestors, whereas independently trained models and distilled students do not. This distinction matters because a distilled student may reproduce a teacher's behavior without inheriting its weights. The resulting method provides a passive, post-hoc provenance signal for white-box residual models without relying on metadata, watermarks, training data, or forward passes.

1 Introduction

The open-weight ecosystem has transformed how neural networks are distributed and reused. When Meta released LLaMA (Touvron et al., 2023), the community produced instruction-tuned variants within weeks. Platforms like Hugging Face now host hundreds of thousands of model checkpoints (Hugging Face, 2024), many of which are fine-tuned (Devlin et al., 2019; Ouyang et al., 2022), quantized (Han et al., 2016; Dettmers et al., 2022), pruned (Han et al., 2015), or merged (Wortsman et al., 2022; Ilharco et al., 2023). This creates a *model supply chain* where downstream checkpoints inherit weights from upstream models, but the lineage is often undocumented or intentionally obscured. Provenance of AI-generated content is now treated as a first-class problem: Google's SynthID watermarks text and images at generation time (Dathathri et al., 2024), yet no analogous mechanism exists for the models themselves. Model reuse is easy; reliable model provenance is not.

Existing provenance mechanisms fall short (Table 1). Cryptographic hashes break after any weight modification. Metadata and model cards (Mitchell et al., 2019) require honest distributors and preserved records; they can be falsified or lost. Watermarks (Uchida et al., 2017; Adi et al., 2018) must be inserted before release; they cannot be applied retroactively to already-deployed checkpoints, and no surveyed scheme proved robust against fine-tuning or pruning (Lukas et al., 2022). Behavioral tests that compare model outputs cannot distinguish weight inheritance from imitation: two models can behave similarly without sharing weights, especially after distillation (Hinton et al., 2015). Activation-based similarity measures like CKA (Kornblith et al., 2019) and decision-boundary fingerprints like IPGuard (Cao et al., 2021) inherit this limitation. We need a passive,



(a) Residual Branch Product (b) Training Induces Diagonal Dominance: ResNet vs PlainNet (c) White-Box Model Lineage Verification

Figure 1: Training leaves traces in residual networks. (a) The residual branch product $M = W_{\text{out}}W_{\text{in}}$ for a block with skip connection; the normalized trace concentration $s(M)$ measures identity alignment. (b) Training dynamics: score matrices $s(i, j)$ from epoch 0 to 200. ResNet (top row) develops strong identity-aligned structure; PlainNet without skip connections (bottom row) remains random. The fingerprint is learned, not an initialization artifact. (c) Lineage verification: the lineage score $\mathcal{L}$ (Eq. 7) aggregates per-block signature similarity. Related checkpoints (fine-tuned, quantized, pruned, LoRA-merged) score high; unrelated checkpoints (independent, distilled) score low, enabling clean separation.

post-hoc signal that reads provenance from the weights themselves.

We ask: given a reference checkpoint and a suspect checkpoint with the same residual architecture (e.g., both LLaMA-2-7B derivatives), can we determine whether the suspect inherited weights from the reference, even after common transformations like fine-tuning, quantization, or pruning? This is distinct from duplicate detection (which hashes solve until the first modification), behavioral similarity (which conflates imitation with inheritance), and watermark verification (which requires forethought). The question is whether shared weight ancestry leaves detectable structure after modification.

Prior work (Park, 2026) observed that correctly paired projections in trained residual blocks develop trace-concentrated products and showed this structure recovers projection correspondences. We build on this observation and ask whether the checkpoint-specific component of these products persists across derived models and can support lineage verification. Our key step is removing the identity-aligned component shared across trained models and comparing centered signatures. Section 5.1 validates the underlying phenomenon through initialization ablations, skip-free controls, and gradient-coupling interventions.

We exploit this structure for provenance verification. Correctly paired projections have high trace-concentration scores; mismatched projections do not. However, independently trained models exhibit the same generic identity-aligned structure, so trace concentration alone does not establish ancestry. We therefore center each branch product and compare checkpoint-specific signatures across models, aggregating similarities into a lineage score (Equation 7). Related checkpoints score near 1; unrelated ones score near 0.

The pairing signal generalizes: across public checkpoints in language, vision, and speech—GPT-2, BERT, LLaMA-2, Mistral, Qwen2.5, DeepSeek-R1, ViT, Whisper, ResNets—architecture-aware products recover projection correspondences with 80–100% accuracy (Table 3). On controlled MLP, ResNet-18, and GPT-2-style language model benchmarks, checkpoints with genuine ancestry (fine-tuning, noise, pruning, quantization, LoRA merging) separate cleanly from independent runs and distilled students (AUROC = 1.000). Behavioral similarity does not fool the method: in our controlled GPT-2 benchmark, independent models trained on the same corpus achieve 77.6% top-1 agreement, and distilled students reach 79%—yet both score $\mathcal{L} \approx 0.002$, correctly rejected as unrelated (Figure 1c). On public LLMs, LLaMA-2-7B derivatives (chat, Vicuna, CodeLlama) score related while OpenLLaMA-7B, trained from scratch, is correctly rejected.

Our contributions:

- **We extend residual-product structure into a checkpoint-comparison framework.** Building on prior observations of trace concentration, we isolate the centered, checkpoint-specific component for lineage verification and validate the phenomenon through controlled interventions.

- **We introduce a centered residual-signature test for white-box lineage verification.** Removing the identity-aligned component common across trained models, we compare checkpoint-specific geometry without training data, forward passes, or watermarks.
- **We evaluate projection pairing and lineage verification at appropriate scopes.** Projection correspondences reach 80–100% accuracy across nine families; controlled MLP, ResNet, and GPT-2-style language model benchmarks with known ancestry evaluate checkpoint-level lineage (AUROC=1.000), complemented by validation on public LLaMA-2 checkpoints.
- **We show the signal survives substantial modification.** It persists through fine-tuning, quantization, pruning, LoRA merging, and RLHF, and tracks partial inheritance in layer grafts and model merges.

2 Problem Setting and Related Work

2.1 White-Box Model Lineage Verification

A verifier receives two checkpoints A and B and determines whether they share weight-level lineage. We assume white-box access and compatible residual architectures (same L and d); cross-architecture comparisons are out of scope. The verifier operates on weights alone: no training data, activations, or forward passes.

Checkpoints are *related* if they share a common weight ancestor through fine-tuning, RLHF, pruning, quantization, or LoRA merging. They are *unrelated* if initialized independently, even when architecture, data, task, or outputs match; a distilled student trained from scratch is unrelated to its teacher, even if it mimics the outputs perfectly. Linear merges induce partial lineage (Appendix F.1).[1] The verifier returns RELATED or UNRELATED. The score is symmetric, $\mathcal{L}(A, B) = \mathcal{L}(B, A)$; directional attribution requires metadata (Appendix D). The method complements cryptographic signing and watermarking where those were never applied.

2.2 Related Work

Table 1 summarizes the comparison.

[1] All appendix references refer to Technical Supplement.

Cryptographic hashes break after any weight modification. Model cards (Mitchell et al., 2019) require honest distributors and can be falsified when checkpoints are redistributed. Proof-of-learning (Jia et al., 2021) requires retained training transcripts that already-released checkpoints lack.

Parameter watermarks (Uchida et al., 2017) and backdoor watermarks (Adi et al., 2018) embed deliberate signals during training. Lukas et al. (2022) found no surveyed scheme that reliably survived fine-tuning and pruning, and all require proactive insertion: unwatermarked legacy checkpoints cannot be verified.

IPGuard (Cao et al., 2021) extracts inputs near classification boundaries; adversarial frontier stitching (Le Merrer et al., 2020) marks decision surfaces with adversarial examples. CKA (Kornblith et al., 2019) and SVCCA (Raghu et al., 2017) compare representations on a probe dataset. All four measure functional similarity and cannot distinguish inheritance from imitation: a distilled student may share decision surfaces while having independent weights. They also require data and forward passes.

Weight cosine, Frobenius distance, and permutation matching (Ainsworth et al., 2023) detect derived checkpoints when the suspect remains close in parameter space, but these are uncalibrated global scalars with no structural account of lineage. HuRef (Zeng et al., 2025) relies on LLM parameter convergence after pretraining. REEF (Zhang et al., 2025) needs probe data or activations. MoTHer (Horwitz et al., 2025) recovers model trees but requires a checkpoint collection rather than a single pair. Our fingerprint is data-free, operates on a single pair against a calibrated null, and localizes inheritance to specific blocks.

Park (2026) observed that correctly paired projections in trained residual blocks produce trace-concentrated products and used this to recover projection correspondences. We build on that observation: rather than treating trace concentration as evidence of ancestry, we remove the identity-aligned component and compare checkpoint-specific signatures. This enables a model-level test of weight inheritance under post-training transformations, reparameterizations, partial inheritance, and distillation.

Method	Retro.	FT	Quant	Distill
Crypto Hash	✓	✗	✗	✗
Metadata/Logs	✗	✓	✓	✓
Watermarking	✗	$\sim$	$\sim$	✗
Behavioral Fingerprints	✓	✓	✓	✗
Ours	✓	✓	✓	✓

Table 1: Comparison with prior provenance methods. Retro.: applicable retroactively. FT: survives fine-tuning. Quant: survives quantization. Distill: correctly classifies distilled copies as unrelated. $\sim$: partial or unreliable.

Architecture	Residual branch $F(x)$	Product M
Transformer MLP / BasicBlock	$W_2\sigma(W_1x)$	W_2W_1
Attention V/O path	$W_O\mathrm{Attn}(x)W_Vx$	W_OW_V
Attention Q/K path	bilinear $x^\top W_QW_K^\top x$	$W_QW_K^\top$
Bottleneck ResNet	$W_3\sigma(W_2\sigma(W_1x))$	$W_3W_2W_1$
SwiGLU MLP	$W_{\mathrm{down}}[\sigma(W_{\mathrm{gate}}x)\odot W_{\mathrm{up}}x]$	$W_{\mathrm{down}}W_{\mathrm{up}}$

Table 2: Architecture-aware residual branch products. Each M composes the linear maps of one branch (dropping nonlinearities) so it maps the residual stream to itself. Factorization must match the architecture; incomplete products destroy the signal (Appendix F.2).

3 Trace Concentration in Residual Branch Products

A residual block computes $x_{\ell+1} = x_\ell + F_\ell(x_\ell)$, where F_ℓ is the residual branch. For a branch with K linear layers $W_1, \ldots, W_K$ interleaved with nonlinearities, the *residual branch product* composes the linear factors, dropping nonlinearities:

$$M_\ell = W_KW_{K-1}\cdots W_1 \in \mathbb{R}^{d\times d} \qquad (1)$$

Individual factors W_k are rectangular and do not share an input–output basis; their composed product maps the residual stream back into its own coordinate space, so its diagonal entries compare like with like. The exact factorization is architecture-dependent (Table 2).

3.1 Why Residual Training Induces Identity Alignment

Under the Frobenius inner product $\langle A, B\rangle = \mathrm{tr}(A^\top B)$, the identity matrix and the traceless matrices span orthogonal subspaces of $\mathbb{R}^{d\times d}$. Any branch product therefore splits into a component along the identity, with coefficient $\langle M_\ell, I\rangle/\langle I, I\rangle = \mathrm{tr}(M_\ell)/d$, and an orthogonal traceless remainder:

$$M_\ell = -\varepsilon_\ell I + E_\ell \qquad (2)$$

where $\varepsilon_\ell = -\mathrm{tr}(M_\ell)/d$ and $\mathrm{tr}(E_\ell) = 0$. Traces are predominantly negative (Appendix E.2), so ε_ℓ is typically positive; the score uses magnitude.

The *normalized trace concentration* (Park, 2026) measures this identity alignment:[2]

$$s(M) = \frac{|\mathrm{tr}(M)|}{\|M\|_F}, \qquad (3)$$

the fraction of the matrix's energy along the identity direction, up to $\sqrt{d}$. Training moves energy into the identity component: correctly paired branch products score far above unpaired or untrained ones (Figure 1b).

The skip connection changes what the branch must learn. A plain layer must transmit the entire representation; a residual branch learns only a correction around an identity path that already transmits everything. This has a gradient consequence: W_{in} and W_{out} are the two ends of one correction, and the loss reaches both through the same branch. ∇W_{out} depends on the branch input, a function of W_{in}; ∇W_{in} depends on the upstream gradient, which flows through W_{out}. Their updates are therefore correlated at every step by construction. In a plain network no such pairing is privileged, every adjacent pair of layers is coupled equally so no per-block signature can form.

The correlation accumulates along the identity direction because the skip means the branch learns a correction, not the full map. Small coordinated corrections to an identity baseline accumulate along the diagonal. Section 5.1 shows that injecting this component directly, without backpropagation, suffices to build the fingerprint.

A natural alternative account is dynamical isometry (Pennington et al., 2017): if training enforced near-orthogonal block Jacobians $J = I + J_F$, identity alignment would follow as a byproduct of pressure toward $J \approx I$. This means trained Jacobians should be more orthogonal than at initialization, which we do not observe in Section 5.1.

3.2 Margin Analysis

By orthogonality of decomposition (2), $\|M_\ell\|_F^2 = \varepsilon_\ell^2 d + \|E_\ell\|_F^2$: the score depends on M only through the ratio of identity to traceless components. For completeness, we restate the resulting correct-pair

[2] Park uses "diagonal-dominance ratio"; we prefer "normalized trace concentration" to avoid confusion with row/column diagonal dominance in matrix analysis.

margin in the notation used by our lineage formulation.

Proposition 1 (Correct-pair score)**.** *Let $M = -\varepsilon I + E$ with $\operatorname{tr}(E) = 0$ and $\varepsilon \neq 0$. Then:*

$$s(M) = \frac{\sqrt{d}}{\sqrt{1 + \|E\|_F^2/(\varepsilon^2 d)}} \leq \sqrt{d} \tag{4}$$

with equality if and only if $E = 0$.

Proof. $|\operatorname{tr}(M)| = |\varepsilon| d$ and $\|M\|_F^2 = \varepsilon^2 d + \|E\|_F^2$ by orthogonality. Substitute into (3).

Geometrically, $s(M)/\sqrt{d}$ is the cosine of the angle between M and the identity. Under the independent-factor null considered for mismatched projections, this yields the corresponding baseline:

Corollary 1 (Mismatched baseline)**.** *For products of projections from independent blocks, $\mathbb{E}[s] \approx 1/\sqrt{d}$.*

The margin scales as d: wider networks separate correct from incorrect pairings more cleanly. Empirically the correct-pair score grows as $d^{0.87}$ (Section 5.2).

4 From Diagonal-Dominance Fingerprints to Lineage Verification

Trace concentration identifies valid trained pairings, not ancestry: independently trained models exhibit the same identity-aligned component. The verifier operates on the centered traceless remainder E_ℓ, using trace concentration only to gate unreliable branches. The identity component is generic; E_ℓ is checkpoint-specific, surviving weight-preserving transformations but not reinitialization.

4.1 Centered Residual Signatures and Block Alignment

For each branch product M_ℓ we retain only the traceless remainder of decomposition (2), normalized to a unit vector:

$$R_\ell = M_\ell - \frac{\operatorname{tr}(M_\ell)}{d} \cdot I, \quad \phi_\ell = \frac{\operatorname{vec}(R_\ell)}{\|\operatorname{vec}(R_\ell)\|_2} \tag{5}$$

$R_\ell = E_\ell$ from (2). The signature ϕ_ℓ carries checkpoint-specific geometry that weight-preserving transformations inherit and independent training cannot reproduce. Because M_ℓ is invariant under hidden permutation and reciprocal rescaling, so is every signature built on it (Section 5.4).

Given reference A and suspect B with L blocks each, we form the similarity matrix $G_{ij} = \langle \phi_i^A, \phi_j^B \rangle$, gated by trace concentration (Appendix D), and recover the block correspondence via the Hungarian algorithm (Kuhn, 1955):

$$\pi^* = \arg\max_{\pi \in \mathcal{P}_L} \sum_i G_{i,\pi(i)} \tag{6}$$

For derived checkpoints preserving block order, π^* should be the identity; recovering it without metadata confirms that signatures track block identity. Complexity: branch products $O(Ld^2h)$, similarity matrix $O(L^2d^2)$, assignment $O(L^3)$.

4.2 Lineage Score and Verification Test

Diagonal dominance certifies trained residual structure, not ancestry: a verifier built on the dominance score alone achieves AUROC=0.417 (Appendix F.3, Figure 7). To distinguish lineage, we compare the centered signatures ϕ_ℓ. The lineage score averages signature similarity over aligned branches:

$$\mathcal{L}(A, B) = \frac{1}{L} \sum_{\ell=1}^{L} \langle \phi_\ell^A, \phi_{\pi^*(\ell)}^B \rangle \tag{7}$$

Each term is a cosine similarity, so $\mathcal{L} \in [-1, 1]$ with $\mathcal{L}(A, A) = 1$. In practice, all observed scores are non-negative: related checkpoints score near 1, unrelated ones near 0, since independently initialized weights produce uncorrelated E_ℓ. The score is symmetric.

We calibrate against a null distribution $\mathcal{N}_A = \{\mathcal{L}(A, U_j)\}$ from independently trained models U_j. The protocol returns RELATED if $\mathcal{L}(A, B)$ exceeds the maximum null score, and UNRELATED otherwise:

$$V(A, B) = \begin{cases} \text{RELATED} & \text{if } \mathcal{L}(A, B) > \max_{U_j} \mathcal{L}(A, U_j) \\ \text{UNRELATED} & \text{otherwise} \end{cases} \tag{8}$$

With n null models, this empirical threshold yields a conformal p-value of at most $1/(n+1)$; no distributional assumptions are required. We report AUROC as the primary metric, which sidesteps threshold selection entirely. Compatibility (same L and d) is checked before scoring; the protocol returns INCOMPATIBLE otherwise. Memory: $O(L^2 + d^2)$ beyond the checkpoints.

5 Experiments

We validate the claims of Sections 3–4: fingerprint existence (Section 5.1), cross-architecture generalization (Section 5.2), lineage verification

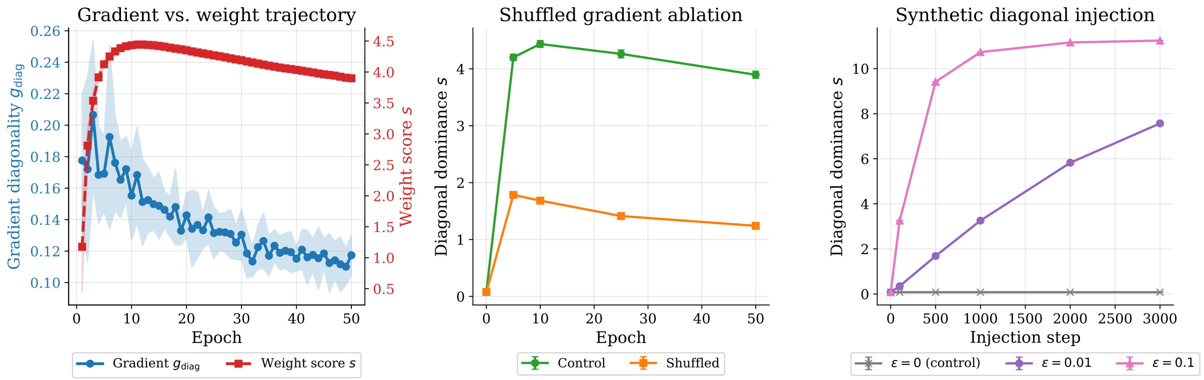


Figure 2: (a) Gradient diagonality g_{diag} (blue) stays flat at $\sim$0.15 while the weight score s (orange) rises to $\sim$4.0: individual gradient updates are not diagonal, but their accumulation is. (b) Shuffling ∇W_{out} across blocks (orange) drops the final score by 68% compared to control (blue), showing that within-block coupling is necessary. (c) Injecting synthetic diagonal updates $\Delta W = -\varepsilon \cdot e_i^\top$ directly into weights builds the fingerprint from scratch without any backpropagation, showing that coupling is sufficient.

(Section 5.3), baseline comparisons (Section 5.4), and survival under post-training modification (Section 5.5).

5.1 The Fingerprint Exists and Is Learned

Section 3.1 predicts that the fingerprint is learned, requires the skip connection, and arises from gradient coupling rather than isometry. We test all three.

We train depth-24 residual MLPs (in_dim=24, hidden=64) on CIFAR-10 (Krizhevsky, 2009) for 200 epochs (3 seeds) under seven initialization schemes: orthogonal, Kaiming-normal/uniform, Xavier-normal/uniform, uniform, and Gaussian σ=0.02. Input-output block pairing accuracy starts at chance ($\leq$2.1%) and reaches 93–100% after training for every scheme that converges (Appendix E.1); the chance-level baseline follows from a null model in which all pairs score uniformly (Appendix C). Orthogonal initialization satisfies dynamical isometry exactly at initialization yet yields zero correct pairs, confirming that the fingerprint requires training.

Using the same architecture, we compare ResNet-24 against PlainNet-24, which removes only the skip connection. ResNet-24 reaches 100% pair accuracy (AUROC 0.98); PlainNet-24, trained to similar loss (1.56 vs. 1.35), stays at chance (3%, AUROC 0.51). Training alone does not produce the fingerprint; the skip connection is required.

Then, we measure the normalized Jacobian deviation $\delta_J^{\text{norm}} = \|J^\top J/\|J\|_F^2 - I/d\|_F$ across the GPT-2 family (small/medium/large/XL), comparing pretrained weights against random initialization. Trained GPT-2 Jacobians are 5–12$\times$ less orthogonal than at initialization (GPT-2-small: 0.297 vs 0.025). The fingerprint grows while orthogonality decays, ruling out isometry (Appendix E.2).

To test whether gradient updates are themselves diagonal, we train 48-block residual MLPs (in_dim=128, hidden=256, CIFAR-10, 50 epochs, 3 seeds) and measure gradient diagonality $g_{\text{diag}} = |\text{tr}(\nabla W_{\text{out}} \nabla W_{\text{in}})|/\| \cdot \|_F$ alongside the weight score $s(M)$. The gradient product stays flat ($\sim$0.15), meaning individual updates are not diagonal. Yet the weight score climbs from 0.08 to $\sim$4.0 (Figure 2a). The diagonal structure emerges from many small correlated updates accumulating over training, not from any single diagonal gradient step.

Two interventions isolate gradient coupling as the cause. *Necessity:* shuffling ∇W_{out} across blocks before each optimizer step (block i's W_{out} receives gradients from a random block j) breaks only the within-block pairing while preserving valid gradients. This cuts the fingerprint by 68% (s: 3.90 $\to$ 1.24) at comparable accuracy (47% vs 54%) (Figure 2b). *Sufficiency:* injecting synthetic updates $\Delta W_{\text{in}} = -\varepsilon \cdot e_i$, $\Delta W_{\text{out}} = -\varepsilon \cdot e_i^\top$ (ε=0.01) directly into weights for 3000 steps, with no loss function or backpropagation, builds the fingerprint from scratch (s: 0.08 $\to$ 7.57) (Figure 2c).

5.2 Architecture-Aware Projection Pairing Across Model Families

We test whether the block alignment procedure of Section 4.1 recovers correct block correspondences across diverse architectures.

Model	Layers	Pair Acc	AUROC
GPT-2 (124M–1.5B)	12–48	**100%**	1.00
BERT-base / ViT-B/16	12	**100%**	0.97–1.00
Mistral-7B / LLaMA-2-7B	32	**100%**	0.96–1.00
LLaMA-2-7B-chat (+RLHF)	32	**100%**	0.96–1.00
Qwen2.5-7B / DeepSeek-R1	28–32	80–**100%**	0.93–1.00
Whisper (tiny/base/small)	4–12	**100%**	0.85–1.00
ResNet-50/101/152	5–35	91–100%	0.96–1.00

Table 3: Cross-architecture projection-pair recovery. Pair accuracy measures whether architecture-aware residual products recover the correct within-checkpoint projection correspondence. This experiment evaluates the underlying pairing signal, not cross-checkpoint lineage verification. Ranges span model sizes within each family. Sub-100% SwiGLU sub-paths are recovered by joint factorization. Random-init baselines: $\leq 9\%$.

Transformation	n	Mean $\mathcal{L}$	Min $\mathcal{L}$
Quantized (INT8/6)	6	0.999	0.999
LoRA merge (rank-8)	3	0.998	0.996
Fine-tuned (1 epoch)	3	0.980	0.980
Pruned (30–70%)	9	0.937	0.855
Distilled student	3	0.002	0.001
Independent	21	0.003	0.000

Table 4: Lineage scores on the held-out test split (3 roots, 45 pairs). Descendants from the 3 test roots are compared against each other and against all 8 roots as independent baselines. AUROC=1.000 with perfect separation.

We extract architecture-aware branch products (Table 2) from 9 model families on Hugging Face (Hugging Face, 2024; OpenAI, 2024) (GPT-2 (Radford et al., 2019), BERT (Devlin et al., 2019), LLaMA-2 (Touvron et al., 2023), Mistral (Jiang et al., 2023), Qwen (Yang et al., 2024), DeepSeek (DeepSeek-AI, 2025) for language; ViT (Dosovitskiy et al., 2021), ResNet (He et al., 2016) for vision; Whisper (Radford et al., 2023) for speech) and apply Hungarian matching on the score matrix $s(i, j)$. Block-pairing accuracy reaches 80–100% across all 9 families, against random-initialization baselines of at most 9% (Table 3). The extraction transfers across GELU (Hendrycks and Gimpel, 2016) and SwiGLU (Shazeer, 2020) activations with no per-family tuning; only the factorization changes.

Extracting MLP branch products $M = W_{\text{proj}}W_{\text{fc}}$ from all four GPT-2 variants (hidden dim d=768/1024/1280/1600), we fit mean correct-pair score $\bar{s}(i, i)$ against dimension. The score grows as $d^{0.87}$ ($4.18 \to 5.46 \to 6.98 \to 7.77$), reaching 15–19% of the theoretical $\sqrt{d}$ bound; score matrices are in Appendix E.2.

5.3 From Fingerprints to Lineage Verification

We evaluate the lineage score $\mathcal{L}$ on depth-24 MLPs (same architecture as Section 5.1). Related checkpoints are generated via fine-tuning, noise, pruning, and quantization; unrelated checkpoints come from independent training and distillation (159 pairs; see Reproducibility Statement). Checkpoints separate (Figure 1c). In Appendix D, Table 10, we observe fine-tune/quant/noise average $\mathcal{L}$=0.99 (min 0.97), pruning averages $\mathcal{L}$=0.81 (min 0.58 at 85% sparsity), while independent/distilled average $\mathcal{L}$=0.08. Extending to ResNet-18 on CIFAR-10, related checkpoints score $\mathcal{L}\geq$0.69 while independent models score $\mathcal{L}<$0.01 (AUROC=1.000; Appendix F.3).

Controlled language model benchmark. To address the gap between synthetic MLPs and uncontrolled public checkpoints, we train 8 GPT-2-style transformers from scratch on TinyStories (Eldan and Li, 2023) with known ancestry. Each root ($\sim$35M parameters: 6 layers, d=384, 50K samples, 3 epochs) is allocated to calibration (n=2), development (n=3), or test (n=3) splits. Per root we generate 7 descendants: 1 fine-tuned (1 epoch continued pretraining), 1 LoRA-merged (rank-8), 3 pruned (30/50/70% sparsity), and 2 quantized (INT8/INT6). We also distill one student per root (T=2.0) from scratch.

Table 4 reports lineage scores on the held-out test split (3 roots). Quantization and LoRA preserve the signal almost perfectly ($\mathcal{L}\geq$0.996); fine-tuning and pruning degrade it but remain well above the null ($\mathcal{L}\geq$0.855). The minimum descendant score (0.855 at 70% sparsity) exceeds the maximum non-descendant score (0.004) by 200×, yielding test AUROC=1.000 (95% CI: 1.0, 1.0). Table 5 compares behavioral similarity against lineage scores: independent models already achieve 77.6% top-1 agreement simply from shared corpus and architecture, while distilled students reach only 79.0%—yet both score $\mathcal{L}\approx$0.002. This confirms that behavioral similarity, whether from shared training setup or explicit distillation, does not imply weight ancestry.

On public LLM checkpoints (Table 6), we compare LLaMA-2-7B-base against 3 known derivatives (chat/RLHF, Vicuna/instruct-tuned, CodeLlama/code-pretrained) and 1 independently trained model (OpenLLaMA-7B), extracting

Comparison	Top-1 (%)	KL	$\mathcal{L}$
Teacher–Distilled	79.0	0.16	0.002
Teacher–Independent	77.6	0.25	0.003

Table 5: Behavioral similarity vs. lineage score. Independent models trained on the same corpus already achieve 77.6% top-1 agreement; distillation adds only 1.4%. Yet both score near zero in lineage, confirming that behavioral similarity—whether from shared training setup or explicit imitation—does not imply weight ancestry.

Pair (base vs.)	Kind	$\mathcal{L}$	W. cos	Verdict
Llama-2-7B-chat	RLHF	**0.995**	0.995	REL.
Vicuna-7B	instruct	**0.996**	0.996	REL.
CodeLlama-7B	code PT	0.336	**0.414**	REL.
OpenLLaMA-7B	scratch†	**4e−4**	0.089	UNR.

Table 6: Public checkpoint lineage validation on the LLaMA-2 family (4 pairs). †OpenLLaMA shares LLaMA-2's exact architecture but is trained from scratch.

SwiGLU branch products $M = W_{\text{down}}W_{\text{up}}$ across all 32 layers (hidden dim d=4096). Three fine-tuned variants score high ($\mathcal{L}$: 0.995, 0.996, 0.336); the independent model scores $\mathcal{L} = 4 \times 10^{-4}$, a $\sim$1000× gap. OpenLLaMA matches LLaMA-2 exactly (d=4096, L=32, d_{ff}=11008) yet was trained from scratch; its UNRELATED verdict can only come from weights. Weight cosine scores higher on CodeLlama, likely because continued pretraining on code shifts the signature more than simple fine-tuning. The score also decays with genealogical distance (Appendix D).

5.4 Lineage Verification Benchmark

We compare against six baselines on a 52-pair MLP benchmark (see Reproducibility Statement for construction details): three data-free weight-space methods (aligned Frobenius, singular-value distance, weight cosine) and three behavioral methods requiring 1024 probe samples each (SVCCA, CKA, and a regression analogue of IPGuard). Table 7 reports aggregate AUROC and Gap-Z = $(\bar{s}_{\text{related}} - \bar{s}_{\text{distilled}})/\sigma_{\text{distilled}}$, the standardized margin by which distilled students fall below related checkpoints. Four weight-space methods, ours included, reach AUROC=1.000 on this benchmark.

High AUROC on unmodified checkpoints could reflect true lineage detection or mere weight similarity. We disentangle them with *function-preserving laundering*: per-block permutation (P) and reciprocal rescaling (D) that leave the network's input–output map exactly unchanged while transforming the weights. Specifically, P permutes hidden units ($W_{\text{in}} \leftarrow PW_{\text{in}}$, $W_{\text{out}} \leftarrow W_{\text{out}}P^{\top}$), and D rescales them ($W_{\text{in}} \leftarrow DW_{\text{in}}$, $W_{\text{out}} \leftarrow W_{\text{out}}D^{-1}$). These transformations leave the branch product $M = W_{\text{out}}W_{\text{in}}$ exactly unchanged ($P^{\top}P = I$, $D^{-1}D = I$), so our signature is invariant by construction. Raw-weight baselines have no such guarantee.

Table 8 shows the results. Our method maintains AUROC=1.000 across all variants—the measured signature deviation is $|\Delta\mathcal{L}| \approx 10^{-8}$, confirming algebraic invariance. Raw-weight baselines collapse: aligned Frobenius drops to 0.495 under P alone and to 0.000 under any D; singular-value distance survives P (spectrum is permutation-invariant) but fails D; weight cosine partially survives both but degrades to 0.805 under PD.

A scale-aware Re-Basin baseline (Ainsworth et al., 2023), which explicitly solves a per-block permutation and scale optimization to undo the laundering, also achieves 1.000 on all variants. The distinction is cost: our signature is invariant with zero alignment work, while Re-Basin must solve L Hungarian assignments plus L least-squares fits for every pair.

We replicate the laundering experiment across four model families to confirm the invariance is not LLaMA-2-specific (Table 9). On each base→fine-tune pair, P permutes $W_{\text{gate}}/W_{\text{up}}$ rows and W_{down} columns; D_{m} (mild rescaling, log-uniform $[0.5, 2]$) scales W_{up} rows against W_{down} columns. Our score is invariant: $\Delta\mathcal{L} < 10^{-7}$ across all families. Weight cosine collapses under P (0.95→0.003) but survives D_{m}. Unrelated controls (OpenLLaMA for LLaMA-2, Mistral for LLaMA-3) stay at the $\leq 10^{-4}$ null regardless of laundering, confirming no false positives.

On layer grafts and linear merges (Appendix F.1), $\mathcal{L}$ tracks the fraction of inherited blocks with Spearman $\rho \approx 0.96$, so the score measures partial overlap rather than merely crossing a threshold. Singular-value distance fails the merge regime (AUROC=0.448), and IPGuard inverts on grafts (AUROC=0.05), ranking higher-overlap suspects as *less* similar.

5.5 Survival Under Post-Training Transforms

Section 5.3 established that the fingerprint detects lineage; here we measure how it degrades as transformation intensity grows. Appendix D,

Method	Type	Data-free	AUROC	Gap-Z ↑	Block Corr.
Centered Residual Signature (ours)	Weight-space	✓	**1.000**	$+53.0$	✓
Aligned Frobenius	Weight-space	✓	**1.000**	$+\mathbf{72.0}$	✓
Singular Value Distance	Weight-space	✓	**1.000**	$+\ 7.9$	✗
Weight Cosine	Weight-space	✓	**1.000**	$+\mathbf{76.3}$	✗
SVCCA (Raghu et al., 2017)	Activation-space	✗	**1.000**	$+45.4$	✗
CKA (Kornblith et al., 2019)	Activation-space	✗	0.829	$+\ 8.6$	✗
IPGuard (Cao et al., 2021) (regression analogue)	Decision-boundary	✗	0.700	$-\ 3.5$	✗

Table 7: Lineage detection baselines (52-pair MLP). Data-free = no probe samples needed. Block Corr. = can identify which blocks match. Centered Residual Signature and aligned Frobenius provide both data-free scoring and block-level correspondences in this benchmark.

Method	none	P	D	PD	PDFT
Ours	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**
Re-Basin+scale	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**
Raw Frobenius	**1.000**	0.495	0.000	0.000	0.000
Raw SVD dist	**1.000**	**1.000**	0.000	0.000	0.000
Raw weight cos	**1.000**	0.856	**1.000**	0.805	0.805

Table 8: AUROC under function-preserving laundering (52 pairs). P=permutation, D=rescaling (log-uniform $[0.1, 10]$), PDFT=P+D then 5 epochs fine-tuning. All variants pass a hard gate ($\max|\Delta f| < 10^{-4}$). Our score is invariant; raw-weight baselines collapse. Re-Basin+scale matches our accuracy but requires L Hungarian assignments + L least-squares fits per pair; ours requires zero alignment.

Base	Pair	Var	$\mathcal{L}$	W.cos
LLaMA-2 7B	chat	P	**.995**	.002
	chat	D_m	**.995**	.949
	OpenLLaMA†	P	$5{\times}10^{-5}$	.000
	OpenLLaMA†	D_m	$5{\times}10^{-5}$	.000
LLaMA-3 8B	Inst	P	**.995**	.003
	Inst	D_m	.995	**.954**
	Mistral†	P	0	.000
	Mistral†	D_m	0	.000
Mistral 7B	Inst	P	**.952**	.003
	Inst	D_m	.952	**.937**
Qwen2.5 7B	Inst	P	**.999**	.003
	Inst	D_m	.999	**.951**

Table 9: Function-preserving laundering on public checkpoints. P=permutation, D_m=mild rescaling (log-uniform $[0.5, 2]$). $\mathcal{L}$ is invariant ($\Delta\mathcal{L} < 10^{-7}$); weight cosine collapses under P but survives D_m. †Unrelated controls stay at $\leq 10^{-4}$ null.

Table 10 covers five transformation families: fine-tuning (same and different target), Gaussian noise ($\sigma_{\text{rel}} \leq 15\%$), magnitude pruning (10–85%), quantization (16–256 levels), and LoRA adapter merging.

Fine-tuning, quantization, and noise leave the signal essentially intact (mean $\mathcal{L} = 0.99$, min 0.97), and LoRA-merged checkpoints score $\mathcal{L} \in [0.97, 0.99]$. Fine-tuning to a different target degrades the score to a mean of 0.94 (min 0.84). Pruning declines most steeply, from $\mathcal{L} = 0.99$ at 10% sparsity to 0.58 at 85%. The real-world extreme of the fine-tuning axis is CodeLlama: after heavy code-specialized continued pretraining it scores $\mathcal{L} = 0.336$, yet remains $\sim$700× above the unrelated null. Across all families, degradation correlates with utility loss ($\rho = -0.83$): no tested transformation removes the signal while preserving the model.

We test adversarial suppression. A white-box adversary optimizes $\min_\Delta \mathcal{L}(A, A + \Delta) + \lambda \cdot \|f_{A+\Delta}(X) - f_A(X)\|^2$, sweeping $\lambda \in \{0, 10^{-2}, 10^{-1}, 1, 10\}$ (Appendix G). With $\lambda \geq 1$, the attack cannot push $\mathcal{L}$ below 0.91. Reaching the unrelated range ($\mathcal{L} \approx 0.08$) requires $\lambda = 10^{-2}$ and costs 12% eval-loss degradation; removing the utility constraint ($\lambda = 0$) destroys the model. The adversary faces the same coupling that makes benign transformations preserve the signal: the fingerprint lives in the correlated component of W_{in} and W_{out}, which encodes the branch's learned function.

6 Conclusion & Ethics Statement

We showed that residual training induces diagonal-dominant structure in branch products—a fingerprint arising from gradient coupling, not initialization or isometry. Centering removes the generic identity component, leaving checkpoint-specific signatures for white-box lineage verification without training data or watermarks. Across nine architecture families the method recovers projection correspondences with 80–100% accuracy and achieves AUROC=1.000 on controlled benchmarks. On public LLMs it confirms LLaMA-2 derivatives while rejecting OpenLLaMA (identical architecture, trained from scratch). The fingerprint survives fine-tuning, pruning, quantization, LoRA merging, and RLHF, and correctly rejects distilled students that mimic outputs without inheriting weights.

Training leaves identifiable traces in residual weight structure—a property that exists whether or not it is measured. Our contribution is formalizing its detection; applications such as provenance verification, IP disputes, and supply-chain auditing follow from the finding itself. The white-box access requirement limits covert use.

Limitations

Scope and architecture constraints. Our method needs white-box access to both checkpoints, so API-only models cannot be verified. It also only works for residual architectures: plain feedforward networks, RNNs, and state-space models are out of scope. Comparisons are further limited to models with matching depth and hidden dimension, so cross-architecture verification (e.g., LLaMA vs. GPT-2, or ViT vs. ResNet) is not supported.

Lineage determination. The lineage score is symmetric, so without external metadata we cannot tell which checkpoint is the ancestor. The score decays monotonically with genealogical distance, but the method compares single pairs and does not reconstruct multi-hop ancestry or family trees. Under partial inheritance (layer grafts, linear merges), the score reflects the fraction of shared blocks but not which blocks were inherited.

Signal degradation under transformation. The fingerprint survives common post-training modifications, but weakens as transformations grow more aggressive. Heavy pruning drops the signal to $\mathcal{L} = 0.58$ at 85% sparsity, and extensive continued pretraining (CodeLlama) lowers it to $\mathcal{L} = 0.336$, though both remain detectable above the null. A white-box adversary can push the score to chance, but only by sacrificing at least 12% utility; no tested attack removes the signal while preserving model function.

Calibration and validation. Verification requires calibrating a null distribution from independently trained models of the same architecture, and the threshold may need adjustment across architecture families. Our controlled GPT-2-style benchmark uses 8 disjoint roots with proper train/dev/test splits, providing valid null-distribution calibration for language models. However, real-world validation covers a limited set of public checkpoints (four LLaMA-2 pairs plus four other model families), so broader ecosystem coverage is untested. SwiGLU architectures (Qwen2.5, DeepSeek-R1) also reach only 80% block-pairing accuracy, and need joint factorization to recover sub-paths reliably. The nine-family result in this paper establishes projection-pair recovery rather than checkpoint-level lineage verification across all nine families.

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A Reproducibility Statement

All results in this paper are produced by deterministic scripts with fixed seeds. The 52-pair MLP lineage benchmark (Tables 10, 7) uses depth-16 MLPs (in_dim=16, hidden=48) and is constructed from 2 trained reference checkpoints, each paired with 5 related-checkpoint kinds $\times 3$ derivative seeds $=$ 30 truly-related pairs (fine-tune, fine-tune to a new target, $\sigma_{\text{rel}} \in [0.01, 0.15]$ Gaussian noise, 10–85% magnitude pruning, and 16–256-level uniform quantization), plus 16 different-seed same-task pairs and 6 distilled-student pairs as unrelated checkpoints. The expanded 159-pair set (Figure 1c) uses 3 depth-24 MLP reference models: each reference is paired with 25 related checkpoints (5 transformation types $\times$ 5 variations) and 28 unrelated checkpoints (15 independent same-task + 5 independent different-task + 5 random-init + 3 distilled), yielding $3 \times 25 = 75$ related pairs and $3 \times 28 = 84$ unrelated pairs ($75 + 84 = 159$ total). Note that these 84 unrelated pairs share only 3 reference model roots; the effective sample size for null calibration is limited by the number of independent roots, not the number of pairs.

The GPT-2-Small-Lite benchmark (Tables 4, 5) uses 8 transformer roots with GPT-2 architecture (L=6 layers, d=384, d_{ff}=1536, 6 heads, $\sim$35M parameters), trained on TinyStories (Eldan and Li, 2023) (50K samples, 3 epochs, batch size 8, lr=3e-4, cosine schedule, fp16, gradient checkpointing). Roots are allocated to calibration (n=2), development (n=3), and test (n=3) splits; Table 4 reports results only on the 3 held-out test roots (21 descendant pairs, 3 distilled pairs, 21 cross-root pairs = 45 total). Each root generates 7 descendants: 1 fine-tuned (1 epoch continued pretraining, lr=1e-4), 1 LoRA-merged (rank-8, α=16, 1 epoch, effective lr=1.25e-5), 3 pruned (magnitude pruning at 30/50/70% sparsity), and 2 quantized (uniform quantization at 256/64 levels, i.e., INT8/INT6). Additionally, 8 distilled students (one per root) are trained from scratch using knowledge distillation (T=2.0, α_{CE}=0.5, α_{KL}=0.5, 2 epochs). Quality metrics for distilled students include top-1 agreement, KL divergence, and perplexity ratio, computed on a held-out validation set.

Exact parameters are listed in Appendix F. Real-architecture pairing (Table 3) loads public HuggingFace checkpoints. Code, configuration files, baseline implementations (aligned Frobenius, singular-value distance, weight cosine, CKA, SVCCA, IPGuard regression analogue), the per-pair score JSONs, and the plotting scripts will be released upon acceptance.

B Use of AI Assistants

The authors bear full responsibility for all scientific claims, experimental results, and analysis in this work. During manuscript preparation, we used Claude to support literature searches, experimental design, code development, and the drafting and editing of text. All AI-assisted material was reviewed, verified, and revised by the authors.

C Extended Theoretical Results

C.1 Null Model (Corollary)

Corollary 2 (Null Model)**.** *For randomly initialized weights with no training, all pairs satisfy* $\mathbb{E}[s(i,j)] = 1/\sqrt{d} + o(1)$ *uniformly in* (i,j)*. Hungarian matching recovers correct pairs at the chance rate* $1/L$*.*

The null model rules out three alternative explanations:

1. Identity alignment is not an artifact of compatible matrix shapes
2. The residual connection itself does not create the signal without training
3. Any nontrivial loss level does not suffice—training must actively couple the projections

The identity-aligned fingerprint is a *learned* property induced by gradient descent under the residual constraint.

D Verification Protocol Details

D.1 Gated Branch Score

Branch similarity is reliable only when both branches exhibit strong trace concentration. We gate the cosine similarity:

$$G(\phi_\ell^A, \phi_m^B) = \langle \phi_\ell^A, \phi_m^B \rangle \cdot \min\left(\frac{s(M_\ell^A)}{\tau_s}, \frac{s(M_m^B)}{\tau_s}, 1\right) \quad (9)$$

where τ_s is the minimum trace concentration across reference branches.

D.2 Statistical Calibration

We calibrate against a null distribution $\mathcal{N}_A = \{\mathcal{L}(A, U_j) : U_j \notin \mathcal{D}(A)\}$ of scores between reference A and independently trained models. Given n null scores, the empirical threshold $\tau = \max_j \mathcal{L}(A, U_j)$ yields a conformal p-value:

$$p(A,B) = \frac{|\{j : \mathcal{L}(A,U_j) \geq \mathcal{L}(A,B)\}| + 1}{n+1} \quad (10)$$

With n null models, the minimum achievable p-value is $1/(n+1)$—roughly 20% with n=4 (LLaMA case study) or 1.2% with n=84 (MLP benchmark). We therefore report AUROC as the primary metric, which aggregates discrimination across all thresholds without requiring specific FPR claims. For the LLaMA-2 case study, we use leave-one-out calibration: when evaluating each independent model, its score is excluded from the null.

Transformation	Mean $\mathcal{L}$	Min $\mathcal{L}$	$>$ max(null)
Fine-tune / Quant / Noise	0.99	0.97	45/45
Fine-tune (diff. target)	0.94	0.84	15/15
Pruning (10–85%)	0.81	0.58	15/15
Independent / Distilled	0.08	0.01	-

Table 10: Lineage score survival under post-training modification (n=75 related, n=84 unrelated). All related checkpoints exceed the maximum null score ($\max \mathcal{L}_{\text{null}} = 0.20$); AUROC=1.000.

E Mechanism Validation

E.1 Initialization Ablation

Table 11 shows pair accuracy before and after training across seven initialization schemes on depth-24 residual MLPs (3 seeds, 200 epochs). All schemes show chance-level accuracy ($\leq$2.1%) at initialization, rising to 93–100% after training. Orthogonal initialization provides dynamical isometry at init yet shows zero correct pairs before training.

Init scheme	Untrained	Trained	AUROC
Orthogonal	0.0%	100%	0.947
Kaiming-normal	2.1%	97%	0.981
Kaiming-uniform	2.1%	98.6%	0.98
Xavier-normal	2.1%	100%	0.990
Xavier-uniform	2.1%	100%	0.99
Uniform	2.1%	93.8%	—
Gaussian σ=0.02	2.1%	13%	0.671

Table 11: Initialization ablation on depth-24 residual MLPs. All schemes show chance-level accuracy before training. The Gaussian σ=0.02 case fails because blocks collapse to near-zero contribution.

E.2 GPT-2 Scaling, Trace, and Jacobian Analysis

Table 12 reports the normalized Jacobian deviation $\delta_J^{\text{norm}} = \|J^\top J/\|J\|_F^2 - I/d\|_F$ across GPT-2 scales. This metric is scale-invariant: any scaled orthogonal matrix scores 0, and larger values indicate non-uniform singular values.

Model	d	Pretrained δ_J^{norm}	Random-init δ_J^{norm}	Ratio
GPT-2-small	768	0.297	0.025	11.7×
GPT-2-medium	1024	0.242	0.026	9.4×
GPT-2-large	1280	0.155	0.025	6.2×
GPT-2-XL	1600	0.122	0.024	5.1×

Table 12: Jacobian orthogonality across GPT-2 scales. Pretrained models are 5–12× less orthogonal than at random initialization, falsifying the dynamical isometry hypothesis. Random-init δ_J^{norm} is nearly constant across scales (≈0.025); pretrained values decrease with model size but never approach the random-init floor.

Figure 3 shows block pairing score matrices $s(i,j)$ for GPT-2 models from 124M to 1.5B parameters.

Figure 4 shows the trace values $\mathrm{tr}(W_{\text{out}} W_{\text{in}})$ per block across all four GPT-2 variants.

E.3 Alternative Methods Comparison

Figure 5 compares block pairing accuracy across GPT-2 scales.

F Extended Benchmarks

F.1 Harder Regime: Layer Grafts and Linear Merges

We construct 40 harder pairs from depth-16 MLPs (same setup as Table 7): (i) **layer grafts** copy $K \in \{0,4,8,12,16\}$ of 16 blocks from reference into an independent donor (related if $K \geq 8$); (ii) **linear merges** compute $w{\cdot}\text{ref}{+}(1{-}w){\cdot}\text{donor}$ with

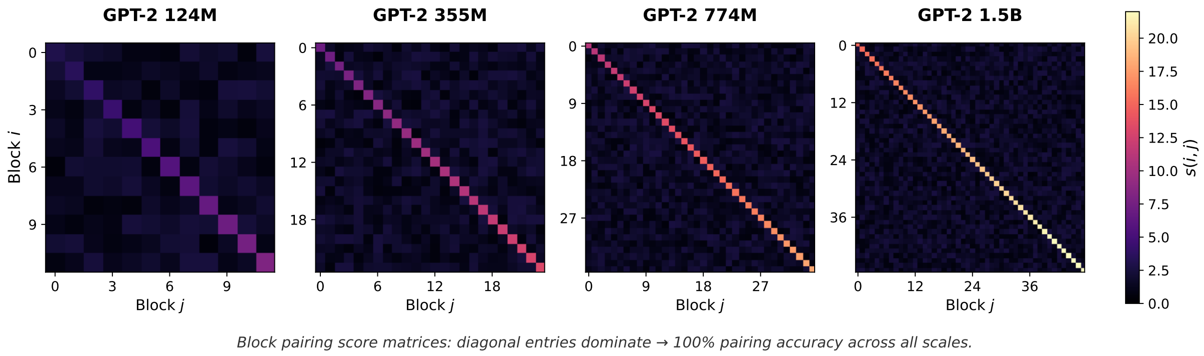

Figure 3: Block pairing score matrices $s(i, j)$ for GPT-2 models from 124M to 1.5B parameters. Diagonal entries dominate, yielding 100% pairing accuracy across all scales.

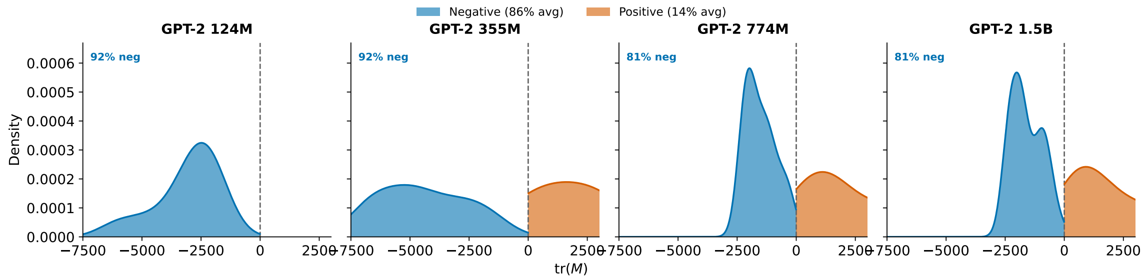

Figure 4: Distribution of trace values $\mathrm{tr}(W_{\mathrm{out}}W_{\mathrm{in}})$ across GPT-2 scales. Blue: negative trace (86% average); orange: positive trace (14% average). The strong skew toward negative values confirms the identity-alignment mechanistic prediction.

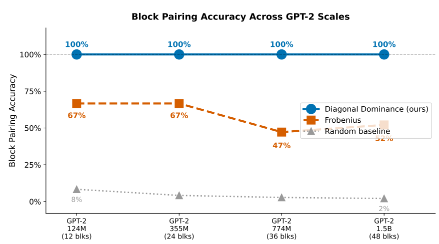

Figure 5: Block pairing accuracy as model size increases. Normalized trace pairing (blue) maintains 100% projection-pair accuracy; Frobenius matching (orange) degrades from 67% to 52%; random baseline (gray) drops from 8% to 2% as the number of blocks increases.

	AUROC		Spearman ρ	
Method	Graft	Merge	Graft	Merge
Centered Residual Signature (ours)	0.979	1.000	+0.96	+0.96
Aligned Frobenius	1.000	1.000	+0.99	+0.99
Weight cosine	1.000	1.000	+0.98	+0.96
Singular-value distance	1.000	0.448	+0.99	+0.23
SVCCA	1.000	1.000	+0.97	+0.99
CKA	0.813	0.781	+0.73	+0.75
IPGuard (regr. analogue)	0.052	0.448	−0.83	+0.22

Table 13: Harder-regime benchmark on layer grafts and linear merges (40 pairs). Related: $K \geq L/2$ for grafts, $w \geq 0.5$ for merges. Weight-space methods (ours, aligned Frobenius, weight cosine) and SVCCA maintain AUROC≈1.000 with $\rho \geq 0.96$. CKA degrades to AUROC≈0.80; IPGuard collapses on grafts (AUROC=0.05, negative ρ); singular-value distance fails merges ($\rho = 0.23$). Our method tracks partial overlap monotonically ($\rho \geq 0.96$), not just binary detection.

$w \in \{0, 0.25, 0.5, 0.75, 1.0\}$ (related if $w \geq 0.5$). We use 2 references and 2 donors per reference. AUROC measures binary related-vs-unrelated detection; Spearman correlation measures whether a method tracks partial overlap monotonically (K/L or w) rather than just clearing a threshold. Table 13 summarizes the 40-pair benchmark.

F.2 ResNet Factorization

On ImageNet-pretrained ResNet-50/101/152 (torchvision), we compare naive two-layer factorization (W_3W_1, skipping the middle conv) against the correct three-layer product ($W_3W_2W_1$) on layer3 bottleneck blocks (n=5/22/35 blocks respectively). Naive W_3W_1 achieves chance-level accuracy; the correct triple product recovers:

- ResNet-50/layer3 (n=5): 100%, AUC 1.000
- ResNet-101/layer3 (n=22): 100%, AUC

0.995

- ResNet-152/layer3 (n=35): 91%, AUC 0.964

F.3 CIFAR-10 ResNet-18 Benchmark

We train two CIFAR-10 ResNet-18 references and three independents. Each reference yields 11 related checkpoints (noise $1-10\%$, pruning $20-80\%$, quantization $16-256$ levels). Results: AUROC=1.000; all related checkpoints exceed the maximum null score. Related checkpoint scores: $\mathcal{L} \in [0.695, 1.000]$; unrelated baseline: $\mathcal{L}_{\max} = 0.004$.

F.4 GPT-2-Small-Lite Benchmark

Figure 6 visualizes the GPT-2-Small-Lite benchmark results from Section 5.3. Panel (a) shows lineage score distributions by transformation type: descendants (quantized, LoRA, fine-tuned, pruned) score high while non-descendants (distilled students, independent roots) score near zero. The minimum descendant score (0.855) exceeds the maximum non-descendant score (0.004) by 200×. Panel (b) shows that distilled students achieve 79% top-1 agreement with their teachers yet score $\mathcal{L} \approx 0.002$, far below any descendant. This validates the paper's central claim: behavioral imitation does not imply weight ancestry.

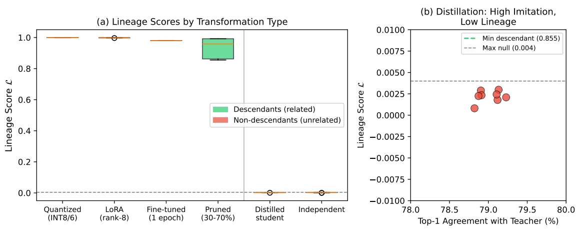

Figure 6: GPT-2-Small-Lite lineage benchmark (8 roots, 120 pairs). (a) Lineage scores by transformation type. Descendants (green) score high; non-descendants (red) score near zero. (b) Distilled students achieve high behavioral agreement (79% top-1) yet score far below the minimum descendant threshold, confirming that imitation $\neq$ inheritance.

F.5 ROC for Lineage Verification

Figure 7 shows that trace concentration alone cannot distinguish related from unrelated trained models.

F.6 Branching Ancestry Recovery

From the MLP benchmark (Table 10 setup), we construct a family tree: root C_1 (trained from scratch), siblings C_2 and C_3 (both fine-tuned from C_1), grandchildren C_4 (from C_2) and C_5 (from C_3). We then measure $\mathcal{L}(C_4, \cdot)$ against all candidates. The lineage score ranks candidates correctly:

$$\begin{aligned} \mathcal{L}(C_4, C_2) = 0.961 \ &> \ \mathcal{L}(C_4, C_1) = 0.910 \\ &> \ \mathcal{L}(C_4, C_3) = 0.867 \ > \ \mathcal{L}(C_4, C_5) = 0.850 \\ &\gg \ \mathcal{L}(C_4, I_k) \approx 0.08. \end{aligned}$$

The score decays monotonically with genealogical distance while maintaining ≈10× family/strangers separation.

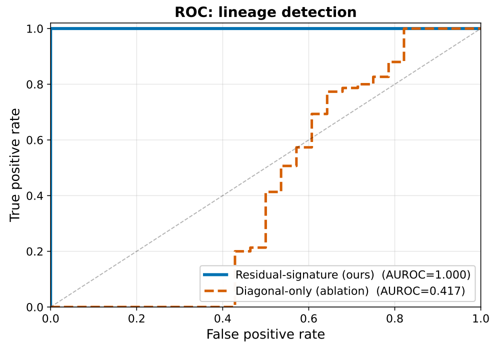

Figure 7: ROC for lineage verification on depth-24 residual MLPs. The residual-signature score achieves AUROC=1.000; trace concentration alone fails (AUROC=0.417) because it cannot distinguish two trained residual models.

G Attack and Robustness Details

G.1 Gradient-Based Suppression Attack

An adversary optimizes perturbation Δ to minimize lineage score subject to utility:

$$\min_{\Delta} \ \mathcal{L}_{\cos}(A, A+\Delta) \ + \ \lambda \cdot \| f_{A+\Delta}(X) - f_A(X) \|_2^2 \qquad (11)$$

λ	Final $\mathcal{L}$	Eval loss	Verdict
0	0.053	39.5	utility destroyed
10^{-2}	0.054	0.77	+12% loss
10^{-1}	0.083	0.70	+1.5% loss
≥ 1	0.91+	0.69	utility preserved

Table 14: Pareto frontier: suppressing $\mathcal{L}$ to chance level requires ≥12% utility loss.